Group 1 — Presentation Script

Modems & RS-232 Interfacing Sequences and Line Coding & Its Techniques

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Course: CIT 3212 — Computer Networks
Topic Area: The Physical Layer (OSI Layer 1)
Companion to the slide deck: “Group 1 Assignment — Modems, RS-232 and Line Coding”

A spoken talk-track: what to say on each slide, plus timing and cues

Suggested total time: 12–15 minutes + Q&A

How to use this script

This document is a **speaker script** for the Group 1 presentation. Each slide in the deck has a matching block below containing:

- **Say** — the words to speak (read naturally, do not rush).
- **Point to / Cue** — what to show or emphasise on the slide.

> **Delivery tips:** split the slides between the two group members; pause after key definitions; keep the waveform and timing diagrams on screen while you explain them; invite questions only at the end unless the lecturer asks.

Suggested timing plan

Section	Slides	Approx. time	Speaker
Opening & Introduction	1–3	1.5 min	A
Modems (definition, operation, types, modulation)	4–9	4 min	A
RS-232 (interface, levels, pin-out, handshake)	10–15	4 min	B
Line coding (definition, schemes, waveforms, comparison)	16–20	3.5 min	B
Summary & close	21–22	1 min	A
Q&A	—	open	Both

Slide-by-slide script

1 Title

~20 s

> **SAY:**
> “Good morning/afternoon. We are Group 1. Our assignment is under the Physical Layer of the OSI model, and we will present two topics: first, **Modems and RS-232 interfacing sequences**; and second, **Line coding and its techniques**. Let’s begin.”

Point to: the title and the course name.

2 Outline

~20 s

> **SAY:**
> “Here is our roadmap. We start with a short introduction, then cover modems, move on to RS-232, then explain line coding and its schemes, and finish with a summary and questions.”

Point to: each numbered item as you mention it.

3 Introduction

~40 s

SAY:
“Both of our topics live in the **Physical Layer — Layer 1** of the OSI model. This is the layer that moves raw bits across a transmission medium; it deals with signals, voltages and timing, not with addresses or packets.

Modems and RS-232 are about *how data is carried between two devices* — the modem turns digital data into a signal for the line, and RS-232 defines the interface that lets a computer talk to that modem. Line coding, on the other hand, is about *how a sequence of bits becomes a signal*. These two ideas are the foundation of everything else.”

Point to: the two cards — “Modems & RS-232” and “Line Coding”.

4 Modems — Definition & Purpose

~40 s

SAY:
“Let’s start with modems. The word **modem** comes from **MOdulator–DEModulator**. Computers produce digital signals, but many transmission lines — such as ordinary telephone lines — carry analogue signals. The modem bridges this gap.

On the sending side it **modulates**: it changes a carrier wave to represent the 0s and 1s. On the receiving side it **demodulates**: it recovers the original bits. Most modems are full-duplex, meaning they do both at the same time. The carrier is simply a sine wave whose amplitude, frequency or phase we can vary.”

Point to: the definition box and the five bullet points.

5 How a Modem Works

~40 s

SAY:
“Here is the flow, step by step. First, bits arrive from the computer over a serial interface such as RS-232. Second, the modulator places those bits onto a carrier. Third, the analogue signal travels over the transmission line. Fourth, at the far end the demodulator recovers the bits. And fifth, those bits are handed to the receiving computer.”

Point to: the numbered steps, one at a time.

6 Modem — Block Diagram

~30 s

SAY:
“This diagram puts it together. On the left we have the computer, or DTE, sending digital data. The modem modulates it, the analogue signal crosses the transmission line, the second modem demodulates it, and the receiving computer gets the original digital data back. Notice that the middle of the chain — the line — carries an analogue signal, while both ends are digital.”

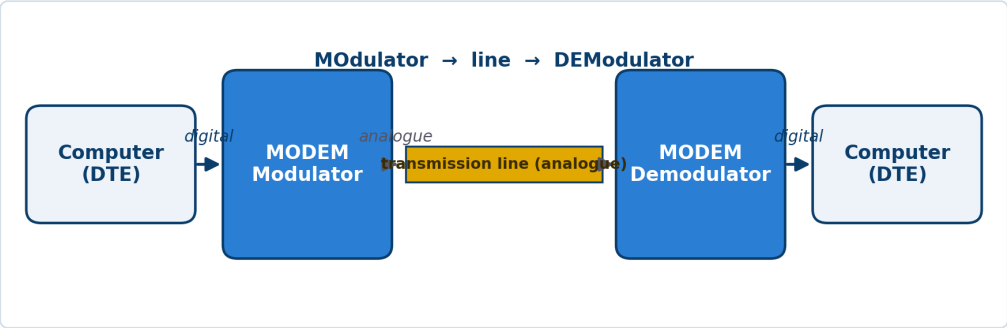


Figure 1 — Modem block diagram (shown on slide 6).

Point to: the digital vs analogue segments of the chain.

7 Types of Modems

~35 s

SAY:
“Modems can be classified in several ways. By direction: half-duplex, which sends both ways but not at once, and full-duplex, which sends both ways simultaneously. By medium: dial-up over phone lines, DSL, and cable modems. By connection: internal cards or external boxes connected by a serial cable. And by carrier: wireless or cellular modems that use radio.”

Point to: the rows of the table.

8 Modulation Techniques

~45 s

SAY:
“Modems use four main techniques. **ASK**, Amplitude Shift Keying, changes the amplitude of the carrier. **FSK**, Frequency Shift Keying, uses different frequencies for 0 and 1. **PSK**, Phase Shift Keying, shifts the phase. And **QAM**, Quadrature Amplitude Modulation, combines amplitude and phase to carry several bits per symbol — which is why it is used in high-speed modems. Remember the relationship: *bit rate equals baud rate times bits per symbol*.”

Point to: the four cards and the formula at the bottom.

9 Modulation Waveforms (ASK / FSK / PSK)

~35 s

> **SAY:**
> “To make this concrete, here are the same data bits — one, zero, one, one, zero, one, zero, one — encoded three ways. In ASK you can see the amplitude switching on and off. In FSK the frequency changes between a low and a high tone. In PSK the phase of the wave shifts. This is exactly what the modem’s modulator is doing.”

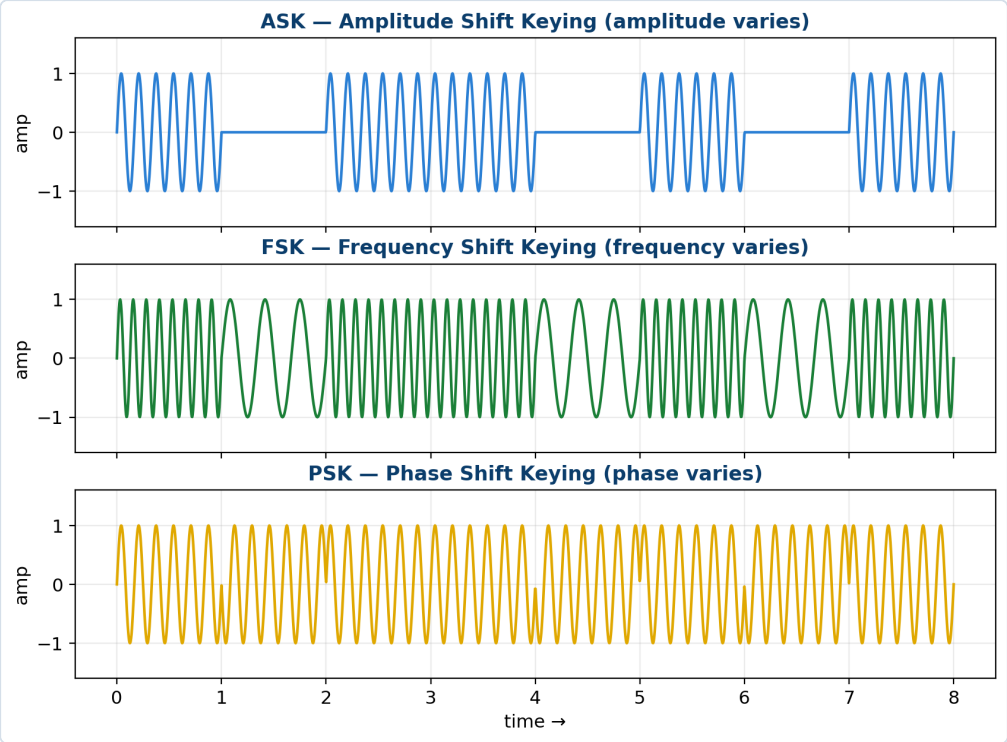


Figure 2 — ASK / FSK / PSK waveforms (shown on slide 9).

Point to: the amplitude change (top), frequency change (middle), phase shift (bottom).

10 RS-232 — What It Is

~40 s

> **SAY:**
> “Now to the second part of the first topic: RS-232. RS-232 is a standard for **serial binary data interchange** — that is, sending data one bit at a time — between two types of equipment. **DTE**, Data Terminal Equipment, is the computer or terminal. **DCE**, Data Communication Equipment, is the modem. The standard defines three things: the electrical characteristics, the mechanical connector, and the procedural sequence of signals.”

Point to: the definition box and the “DTE vs DCE” card.

11 RS-232 — Electrical Characteristics

~35 s

SAY:
“RS-232 uses **negative logic** with large voltage swings to resist noise. A logic 0, called a space, is a positive voltage of plus three to plus fifteen volts — typically plus twelve. A logic 1, called a mark, is minus three to minus fifteen volts. Voltages between minus three and plus three are undefined and treated as noise.

Because it is single-ended, the cable length is limited to about fifteen metres, and speeds are relatively low — up to twenty kilobits per second.”

Point to: the voltage table, then the bullet list.

12 RS-232 — Pin-Out (DB-9)

~45 s

SAY:
“RS-232 uses a DB-9 or DB-25 connector. The key signals are: pin 2, RXD, receive data; pin 3, TXD, transmit data; pin 5, ground. Then the control signals: DTR and DSR tell each side the other is ready, and RTS and CTS are used for flow control. DCD means the modem has detected a carrier, and RI — ring indicator — signals an incoming call. Notice that TXD and RXD are crossed between the two ends, which is why a null modem cable simply crosses these wires.”

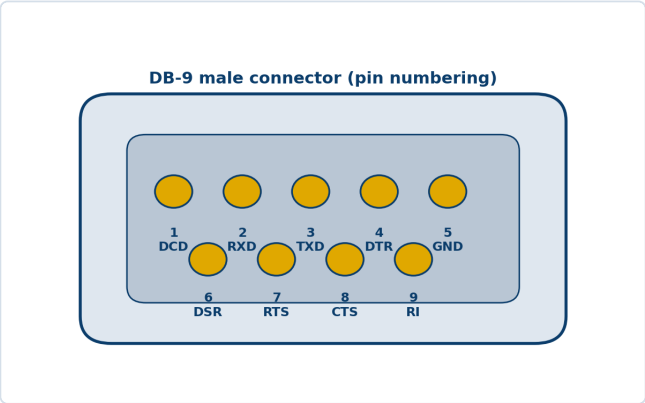


Figure 3 — DB-9 connector pin numbering (shown on slide 12).

Point to: the three signal pairs: TXD/RXD, DTR/DSR, RTS/CTS.

13 RS-232 Handshaking Sequence

~45 s

> **SAY:**
> “Before any data flows, the two ends must ‘shake hands’. First the computer asserts DTR — it is ready. The modem replies with DSR. The computer then asserts RTS, asking to send. The modem answers with CTS, giving permission. Now data flows on TXD and RXD. When the computer finishes, it drops RTS, the modem drops CTS, and finally both drop DTR and DSR to release the connection.
>
> For an incoming call, the modem first pulses RI, the computer raises DTR, the modem answers and raises DCD, and then data can flow.”

Point to: the numbered steps; mention the RI/DCD case for incoming calls.

14 Handshake — Timing Diagram

~30 s

> **SAY:**
> “Here is the same sequence as a timing diagram. You can clearly see the order: DTR and DSR go high first, then RTS and CTS, and only then does data appear on TXD. When RTS and CTS drop, the transfer stops. This ordering is the ‘interfacing sequence’ the assignment asked about.”

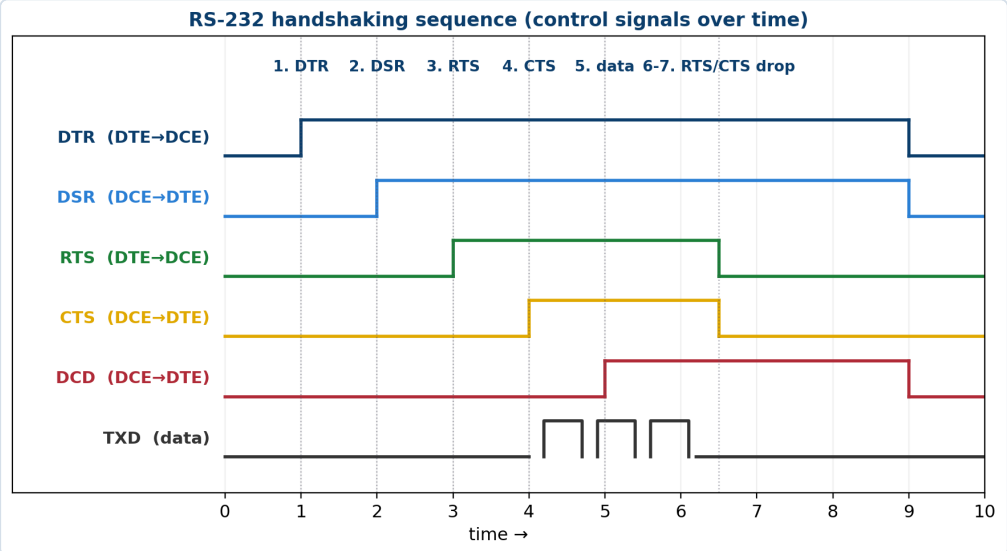


Figure 4 — RS-232 handshake timing (shown on slide 14).

Point to: the sequence of vertical transitions.

15 Flow Control, Null Modem & Limitations

~35 s

SAY:
“Two points to add. First, flow control: hardware flow control uses RTS and CTS to pause the sender, while software flow control uses special characters, XON and XOFF. Second, the null modem: if you want to connect two computers directly — two DTEs — you use a cable that crosses TXD with RXD and RTS with CTS. That is why it is called a ‘null’ modem — no modem is needed.

Finally, the limitations: RS-232 is short-range, fairly slow, point-to-point, and has largely been replaced by USB and RS-485 — though it is still common in industrial and embedded equipment.”

Point to: the three cards in order.

16 Line Coding — Definition & Goals

~40 s

SAY:
“Now the second topic: line coding. **Line coding** is the process of converting a sequence of bits into a digital signal that can be transmitted over a baseband channel — in other words, mapping 1s and 0s to voltage levels on the line. It happens before carrier modulation.

A good line code should avoid a DC component so the baseline does not drift, be self-synchronising so the receiver can recover the clock, resist noise, and use bandwidth efficiently.”

Point to: the definition box and the goals table.

17 Line Coding Schemes

~40 s

SAY:
“Line coding schemes fall into families. **Unipolar** uses levels on one side of the axis — simple, but with a DC component. **Polar** uses both positive and negative voltages; this includes NRZ-L, where the level equals the bit, NRZ-I, where a 1 causes a transition, and Manchester, where there is always a transition in the middle of the bit. **Bipolar** uses three levels — in AMI, zeros are zero volts and ones alternate polarity. And finally, **block coding** such as 4B/5B, and **scrambling** such as B8ZS, are used to guarantee transitions over long distances.”

Point to: the four cards.

18 Line Coding Waveforms

~40 s

> **SAY:**
> “This is the heart of the topic — the same eight bits, one zero one one zero zero zero one, encoded four ways. In NRZ-L the level simply follows the bit: high for 1, low for 0. In NRZ-I every 1 causes a transition and a 0 causes none. In Manchester there is always a transition in the middle of each bit, which gives excellent synchronisation — that is why classic Ethernet used it. And in AMI, zeros are zero volts while ones alternate between positive and negative. Notice how each scheme makes a different trade-off between simplicity and synchronisation.”

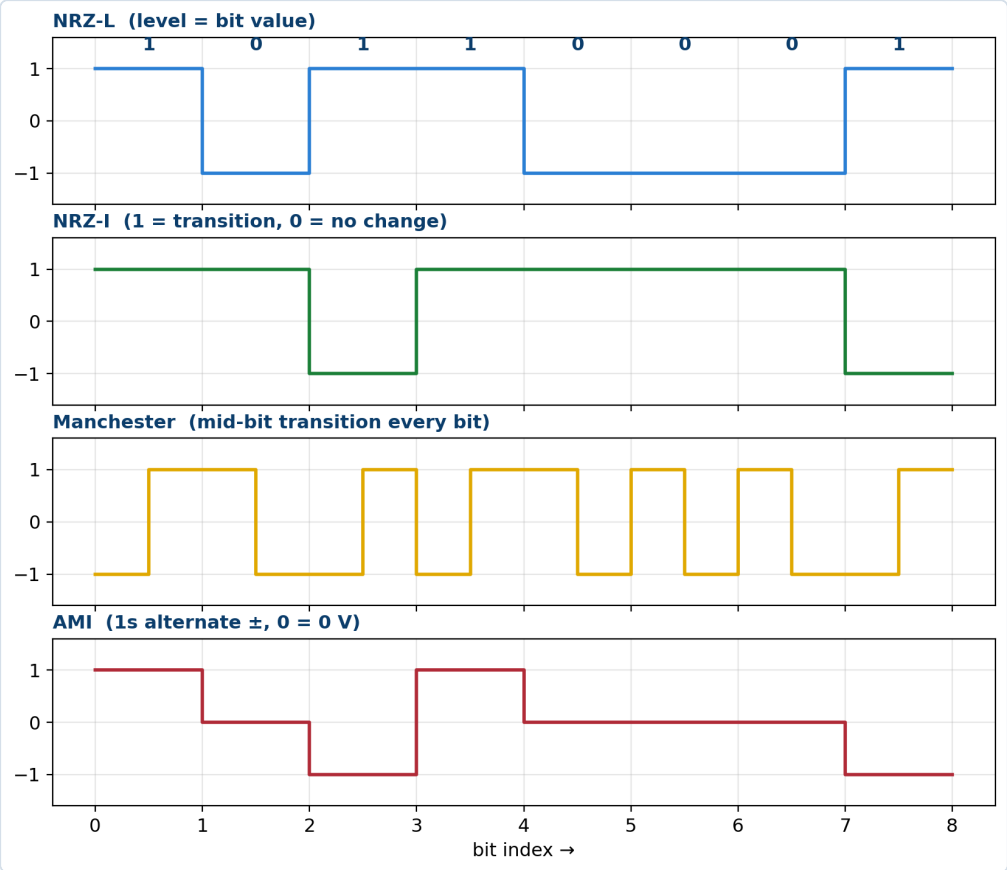


Figure 5 — NRZ-L, NRZ-I, Manchester and AMI (shown on slide 18).

Point to: each waveform in turn; highlight the mid-bit transitions in Manchester.

19 Worked Example

~30 s

> **SAY:**
> “Here is the same data written out numerically so you can check it. For NRZ-L, plus and minus follow the bits. For NRZ-I, each 1 flips the level. Manchester always has a mid-bit transition, and AMI alternates the polarity on each 1 while a 0 stays at zero volts. If you can reproduce this table, you understand the schemes.”

Point to: each row of the code block.

20 Comparison of Line Coding Schemes

~30 s

> **SAY:**
> “This table summarises everything. Look at three columns: does the code have a DC component, is it self-synchronising, and how much bandwidth does it use? Manchester and differential Manchester have no DC and are self-synchronising, but use more bandwidth — which is why they suit LANs. AMI saves bandwidth for long-distance T1 lines. And block coding like 4B/5B combined with NRZ-I is used in Fast Ethernet over fibre.”

Point to: the DC, Self-sync and Bandwidth columns.

21 Summary

~30 s

> **SAY:**
> “To summarise: a modem converts between digital and analogue signals using ASK, FSK, PSK or QAM. RS-232 is the serial interface that connects a computer to a modem, defined by its voltage levels, its pin-out and its handshaking sequence. And line coding converts bits into a digital signal — the best codes avoid DC, stay synchronised and resist noise.”

Point to: the three summary cards.

22 Thank You

~10 s

> **SAY:**
> “That concludes our presentation. Thank you for listening — we are happy to take any questions.”

Point to: the closing slide.

Anticipated questions & model answers

Q1. What is the difference between a modem and a codec?
A modem converts between *digital* and *analogue* signals for transmission. A codec converts between analogue signals and digital *data* for storage or processing (e.g. audio). They are related but serve different purposes.

Q2. Why is RS-232 limited to about 15 metres?
Because it is single-ended — each signal is measured against a common ground, so noise and ground differences build up quickly. Differential standards like RS-485 cancel this noise and reach much further.

Q3. Why does Manchester coding use more bandwidth?
Because it guarantees a transition in every bit, effectively doubling the signal rate compared with NRZ. That extra bandwidth buys reliable clock recovery.

Q4. Why is a DC component a problem?
A steady DC level cannot pass through transformers or AC-coupled equipment, and it causes baseline drift, making the receiver misjudge the voltage threshold and misread bits.

Q5. What is the difference between bit rate and baud rate?
Baud rate is the number of signal changes (symbols) per second; bit rate is the number of bits per second. They are equal only when one symbol carries exactly one bit — as with QAM, one symbol can carry several bits, so the bit rate is higher.

Q6. When would you use a null modem?
When connecting two computers (both DTEs) directly without a modem — for example to transfer files between two PCs over a serial cable.

Companion script to the Group 1 slide deck — CIT 3212 Computer Networks. Speak naturally; the wording above is a guide, not a text to be read word-for-word.